

MODULES COVERED

- ◆ Document Processing
- ◆ Chunking Module
- ◆ Embedding Module
- ◆ Vector Storage
- ◆ Retrieval Module
- ◆ LLM + Prompt
- ◆ HITL Store

Low-Level Design

Module Architecture · Data Structures
Workflow Graph · Error Handling

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1 Module-Level Design

Document Processing ingestion.py		
Function	Description	Returns
<code>load_pdf_documents(pdf_path: Path)</code>	Load PDF pages into LangChain Document objects.	<code>List[Document]</code>
<code>normalize_text(raw_text: str)</code>	Collapse noisy whitespace and normalize page text.	<code>str</code>

Chunking ingestion.py		
Function	Description	Returns
<code>split_documents_into_chunks(documents, chunk_size, chunk_overlap)</code>	Split docs into overlapping chunks with chunk_index metadata.	<code>List[Document]</code>

Embedding vector_store.py		
Function	Description	Returns
<code>build_embedding_model(config: AppConfig)</code>	Construct HuggingFace sentence-transformer model, normalized.	<code>HuggingFaceEmbeddings</code>

Vector Storage vector_store.py / service.py		
Function	Description	Returns
<code>build_vector_store(config, embedding_model)</code>	Create and configure Chroma collection.	<code>Chroma</code>
<code>ingest_documents_into_vector_store(config, chunked_documents)</code>	Embed and persist chunks to Chroma.	<code>Chroma</code>
<code>get_vector_store(config)</code>	Return reusable vector store handle at runtime.	<code>Chroma</code>

Retrieval retriever.py		
Function	Description	Returns
<code>retrieve_relevant_documents(vector_store, query, top_k)</code>	Retrieve top-k chunks with relevance scores.	<code>RetrievalResult</code>
<code>is_query_ambiguous(cleaned_query: str)</code>	Detect ambiguous short or heuristic-matched queries.	<code>bool</code>

LLM / Prompt llm_service.py / workflow.py		
Function	Description	Returns

<code>build_llm(config)</code>	Instantiate ChatGroq with configured model.	ChatGroq
<code>generate_grounded_answer(llm, user_query, retrieved_documents)</code>	Build grounded prompt and invoke LLM.	str

HITL hitl_store.py / streamlit_app.py		
Function	Description	Returns
<code>create_ticket(...)</code>	Create and persist a new escalation ticket.	Dict[str, Any]
<code>list_tickets(...)</code>	Return all tickets filtered by status.	List[Dict]
<code>resolve_ticket(...)</code>	Mark ticket resolved with agent response and timestamp.	Dict[str, Any]

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Data Structures

Document Representation

Type: LangChain Document

Field	Type / Value
<code>page_content</code>	<code>str</code> – normalized chunk text
<code>metadata.source</code>	<code>str</code> – original PDF file path
<code>metadata.page</code>	<code>int</code> – source page number
<code>metadata.chunk_index</code>	<code>int</code> – sequence index after splitting

Query-Response Schema

Type: Returned by workflow per query

Field	Type / Value
<code>answer</code>	<code>str</code>
<code>confidence</code>	<code>float</code>
<code>needs_escalation</code>	<code>bool</code>
<code>escalation_reason</code>	<code>str</code>
<code>retrieved_documents</code>	<code>List[Document]</code>
<code>similarity_scores</code>	<code>List[float]</code>
<code>error</code>	<code>str</code>

HITL Ticket Schema

Type: `hitl_store.py` JSON record

Field	Type / Value
<code>ticket_id</code>	<code>str</code>
<code>session_id</code>	<code>str</code>
<code>user_query</code>	<code>str</code>
<code>escalation_reason</code>	<code>str</code>
<code>confidence</code>	<code>float</code>
<code>retrieved_context</code>	<code>List[Dict[str, Any]]</code>
<code>status</code>	<code>"pending" "resolved"</code>
<code>human_response</code>	<code>str</code>

resolved_by	str
created_at	str (ISO timestamp)
resolved_at	str (ISO timestamp)

Graph State Object

Type: SupportAssistantState (TypedDict)

Field	Type / Value
query	str
cleaned_query	str
retrieved_documents	List[Document]
similarity_scores	List[float]
answer	str
confidence	float
needs_escalation	bool
escalation_reason	str
error	str

3 Workflow Design (LangGraph)

Node Definitions

input_processing_node

- Trim and normalize whitespace in query.
- Validate non-empty input.
- Set early escalation for invalid input (empty or whitespace-only).

retrieve_and_answer_node

- Retrieve top-k documents via similarity search.
- Build LLM prompt from retrieved context chunks.
- Generate grounded answer or detect insufficient context.
- Capture and store any LLM runtime failures.

decision_node

- Inspect retrieval result quality (document count + scores).
- Evaluate ambiguity heuristics on cleaned query.
- Check for runtime error flag.
- Decide: route to output_node or escalation_node.

output_node

- Return final answer for direct-response cases.
- No mutations to state — passthrough to END.

escalation_node

- Convert escalation_reason into user-facing handoff message.
- Set answer field to the escalation response text.

Edge Topology

From	To	Condition
START	input_processing_node	Always
input_processing_node	retrieve_and_answer_node	Always
retrieve_and_answer_node	decision_node	Always
decision_node	output_node	needs_escalation == False
decision_node	escalation_node	needs_escalation == True

output_node	END	Always
escalation_node	END	Always

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Conditional Routing Logic

Direct Answer Criteria

- Query is valid (non-empty after normalization).
- At least one document retrieved from Chroma.
- Confidence (max similarity score) $\geq$ SIMILARITY_THRESHOLD (default: 0.30).
- Query is not flagged ambiguous by heuristics.
- No internal runtime error has occurred.

Escalation Criteria

Trigger	Condition	Reason String
Invalid Input	query is empty or whitespace	"invalid_input"
Missing Context	retrieved_documents is empty	"no_relevant_documents"
Low Confidence	max_score < 0.30 (default threshold)	"low_confidence"
Ambiguous Query	len(tokens) < 4 OR query in heuristic list	"ambiguous_query"
Runtime Failure	LLM invocation raises exception	"llm_failure"

Ambiguity Heuristic Keyword List

"help"	"issue"	"problem"
"not working"	"what should i do"	"can you assist"

5 HITL Design

When Escalation Is Triggered

- User input is invalid (empty query).
- Retrieval yields no relevant chunks from the vector store.
- Confidence score falls below the configured threshold.
- User query matches the ambiguity heuristic.
- LLM service call fails at runtime.

Post-Escalation Flow

#	Actor	Action
1	LangGraph	Sets needs_escalation=True and records escalation_reason in state.
2	escalation_node	Returns user-facing handoff message as answer text.
3	Streamlit	Creates ticket record in data/hitl_tickets.json with full context.
4	Customer Session	User remains in same chat session — no redirect.
5	Agent View	Lists pending tickets; agent sees query, reason, confidence, context.
6	Agent Submit	Agent submits resolution; ticket status → resolved.
7	Delivery	Customer session polls store and injects human response into chat.

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API / Interface Design

CLI Interfaces

Mode	Command
Single Query	
Interactive Chat	
Run Ingestion	

CLI Output Format

CODE

```
Answer: <generated answer text>
confidence=0.82, escalated=false

# Escalated output:
Answer: Your query has been escalated to a human agent.
confidence=0.18, escalated=true
```

Streamlit UI Components

Component	Description
Customer View	Free-text chat input via st.chat_input. Renders answer, confidence badge, escalation status, retrieved context.
Agent View	Lists pending tickets. Shows query, reason, confidence, retrieved snippets. Submit field for human resolution.
PDF Upload Sidebar	Allows live ingestion of a new PDF file. Triggers ingest pipeline on upload.
Ticket Identifier Display	Shows ticket ID and resolved human response once agent has processed the case.

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Error Handling

Error	Location	Cause	Handling
FileNotFoundError	<code>ingestion.py</code>	PDF path does not exist.	Raised immediately in <code>load_pdf_documents</code> . User receives clear path error.
Empty Query	<code>input_processing_node</code>	Query is empty or whitespace after strip.	Trapped in graph node. Escalation triggered with reason 'invalid_input'.
No Chunks Retrieved	<code>decision_node</code>	Similarity search returns empty list.	Decision node escalates with reason 'no_relevant_documents'.
LLM Failure	<code>retrieve_and_answer_node</code>	ChatGroq raises an exception at inference time.	Exception caught, error stored in state. User receives escalation, not stack trace.
Invalid Chunk Config	<code>ingestion.py</code>	<code>chunk_size</code> or <code>chunk_overlap</code> fails validation.	<code>ValueError</code> raised before ingestion proceeds.
Missing Vector Store	<code>service.py</code> / runtime	Ingestion not run before query is attempted.	Retrieval or store initialization fails upstream with informative error.
Missing Env Variables	<code>AppConfig</code>	Required environment variables not set.	Caught by pydantic-settings validation on startup.

Operational Limitations

- Vector store assumed to exist at query time — no lazy-init fallback.
- HITL persistence is local JSON — not suitable for multi-process or distributed deployments.
- No retry logic on LLM failures — single failure triggers escalation immediately.